Bayesian nonparametric glottal inverse filtering

The problem

Speech is locally the convolution of the *glottal wave* and the *vocal tract filter*.

We can measure speech directly, but not its constituents.

Computing them from speech alone is severely underdetermined. This is *glottal inverse filtering*, open for 50+ years.

Beyond fundamental speech science, it matters for early diagnosis of neurodegenerative diseases like Parkinson's and Alzheimer's, which interfere with speech production early on, and for biomarking of heart failure.

Our solution

We recast the problem as Bayesian inference: *Bayesian nonparametric glottal inverse filtering* (BNGIF).

- A Gaussian process prior over glottal waves, learned from a large corpus of expensive physics-based simulations.
- The corpus is embedded in a small latent space and modelled as a Gaussian mixture: a mixture of cheap surrogate models.
- The surrogates are the prior of an efficient variational inference method. State of the art on the EGIFA/speech benchmark.

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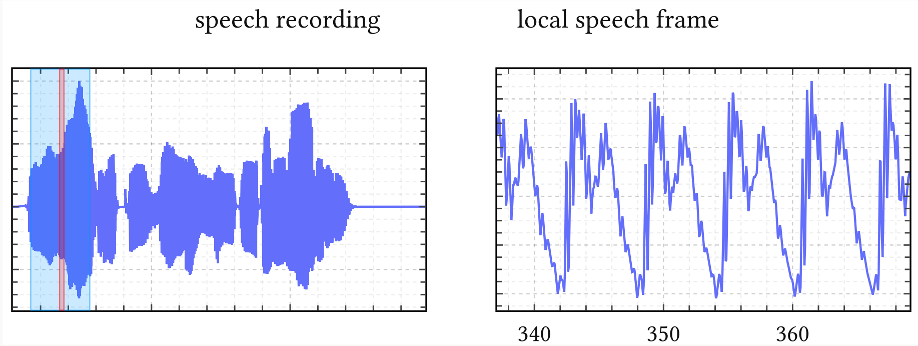
Speech signal



Speech diagnostics

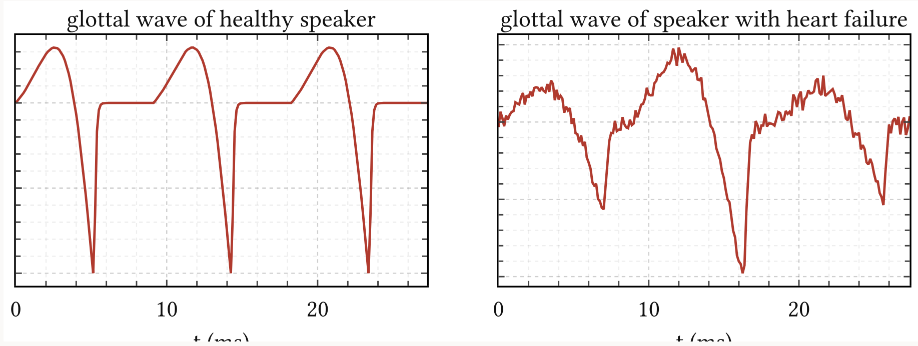
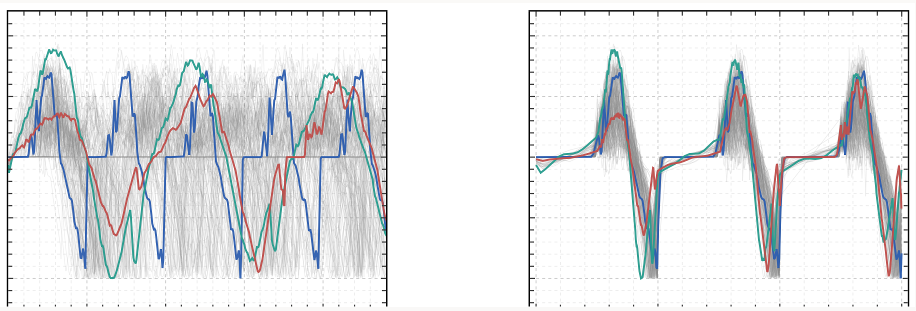
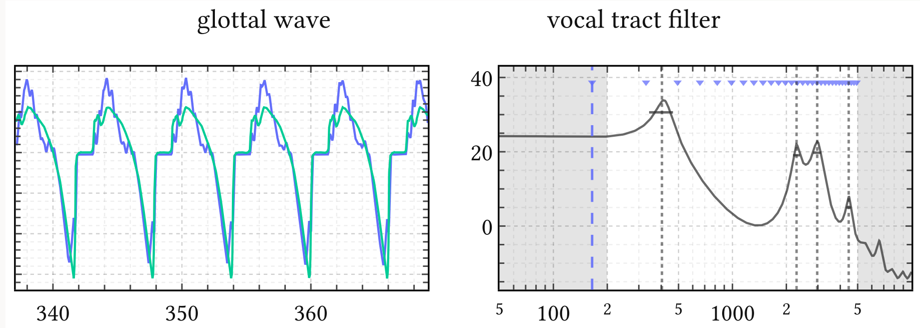


illustration: modal vs perturbed phonation

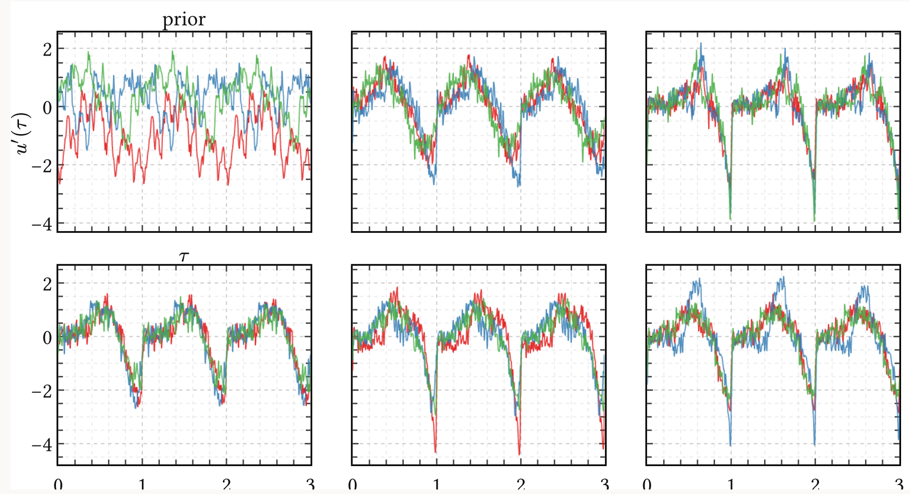


large corpus of glottal waves, registered cycle by cycle

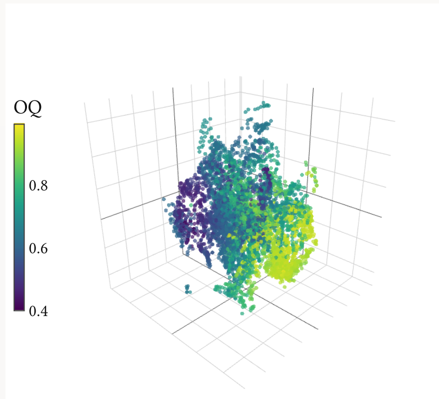
Glottal wave & vocal tract filter



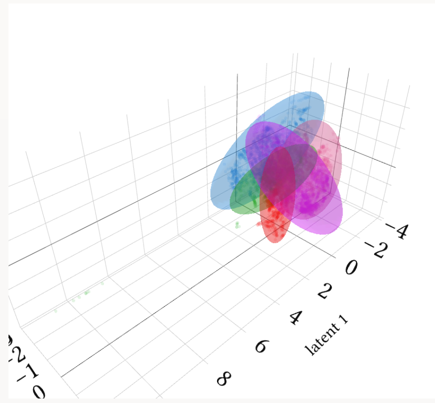
Corpus and mixture model



archetypes of the corpus, with samples



embedded corpus; each point is a glottal wave



mixture model of the embedded corpus